

NORMA-O Requirements Specification Document	
1. Purpose	
NORMA-O is an OWL 2 DL ontology for representing legal provisions as formal norm instances. Each instance captures both the normative content (deontic type, legal agent, regulated action, legal object, and source provision) and the extraction context, including who produced the annotation, by what method, with what confidence, and under what review status.	
2. Scope	
NORMA-O covers legal norms extracted from regulatory provisions, including their deontic type, structural arguments, and conditional applicability. It also represents the annotation context that produced each norm instance: annotator, extraction method, confidence, and review status	
3. Implementation Language	
OWL 2, serialised in Turtle.	
4. Intended End-Users	
U1. Domain experts involved in regulatory compliance work.	
5. Intended Uses	
Use 1. Formal schema for representing and querying legal norm instances. Use 2. Vocabulary for deriving applicable norms via SWRL rules under factual conditions. Use 3. Foundation for SPARQL queries over regulatory annotations. Use 4. Reusable model applicable to any regulatory domain.	
6. Ontology Requirements	
6a. Non-Functional Requirements	
NFR1. FAIR compliant NFR2. Consistent under OWL 2; no unsatisfiable classes. NFR3. Extensible without modifying existing definitions.	
6b. Functional Requirements — Competency Questions (aligned with Section 7 of the paper)	
CQG1. Normative content retrieval	CQG2. Source traceability and provenance
CQ1. Which norms apply to a given legal agent?	CQ7. What annotator, extraction method, and confidence score are recorded for each norm?
CQ2. Which norms concern a given legal object?	CQ8. Which norms carry a given review status?
CQ3. Which norms regulate a given legal action?	CQ9. Which norms carry a given binding force, norm status, or compliance criticality?
CQ4. Which norms originate from a given regulation, article, or paragraph?	CQ10. Which norms are classified as critical or high compliance priority?
CQ5. Which source text is linked to a given norm?	CQ11. Which norms are derivable given a factual scenario? (Answered via SWRL inference; requires a compatible reasoner.)
CQ6. Which norms share the same legal source?	

Table 1: NORMA-O Requirements Specification Document, following the LOT methodology template. <https://w3id.org/def/norma-o>.